

Intensity Estimation of Spectroscopic Signals with an Improved Sparse Reconstruction Algorithm

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Abstract—One of the main measurements performed in a nuclear spectroscopy experiment is the activity of an unknown radioactive source. The use of digital apparatus and physical perturbations, known as pile-up effect, make this measurement difficult when the activity of the source is high. In recent contributions, the use of compressive sensing methods yielded good estimates of this activity. This paper presents an improvement of a previously described method. It takes into account the fact that the signal used for the activity estimation is sampled, and introduces another plug-in estimation to counterbalance the bias introduced by the sampling. Results on simulations and real data validate the proposed approach, but illustrate that a good fit between the dictionary used and the signal at hand is required.

Index Terms—Sparse Reconstruction, Point processes, Poisson Processes, LASSO, Compressive Sensing, Nuclear Spectroscopy

I. INTRODUCTION

In nuclear spectrometry experiments, particles interact with a detector at random times, creating electrical pulses which are afterwards analyzed [1]. Among other things, the practitioner is usually interested in measuring the activity of radioactive sources. However, when the activity of the radioactive source is high, generated pulses may overlap. This phenomenon is known in the literature as pile-up effect. An example of a sampled electrical signal stemming from a radioactive source is displayed in Figure 1. It can be understood that a basic threshold on such a signal in order to detect the photon arrival times yields a severe underestimation of the activity.

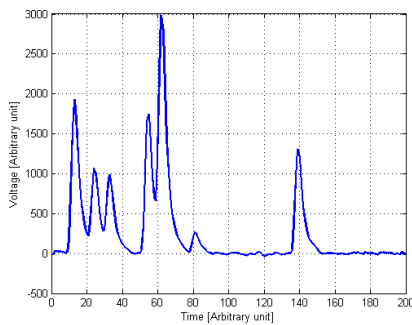


Fig. 1: Example of electrical signal. The high activity makes the intensity estimation difficult.

The problem of activity estimation has been extensively studied in the field of nuclear instrumentation since the 1960's,

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and has received considerable attention in various fields (see [2]–[8]) since then. Based on recent developments in compressive sensing, new methods for counting rate estimation from the sampled recorded electrical signal were proposed in [9]–[11]. More specifically, [11] derived theoretical bounds which showing that a good estimate of the activity can be obtained, based on a thresholded, sparse regressor of the electrical signals. However, it has been noted in [11] that, for very high levels of activity, this estimator tends to underestimate the true activity. The main explanation of this phenomenon is that our observations are sampled signals, and that the estimator [11] can only infer the arrival times from the sampling points. Thus, as the activity increases, the probability increases that two or more photons interact with the detector between two sampling points.

The purpose of this paper is to correct the underestimation of the estimate [11] by means of known results from point process theory. More specifically, we suggest not to estimate the arrival times from the LASSO regressor in a straightforward manner, but rather to estimate, first, the probability that one or more photons arrived between two sampling times from the LASSO regressor. This estimate is then plugged in a non-linear relation between this probability and the actual activity. Section II describes the novel estimator suggested, and section III presents the results obtained in simulations. A example of application on a real dataset is also presented.

II. ACTIVITY ESTIMATION BASED ON A LASSO RECONSTRUCTION

A. Model Description

We observe a signal on the time interval $[0; T]$ uniformly sampled on $\mathcal{T} = \{0 = t_0, t_1, t_2, \dots, t_{N-1} = T\}$ with sampling period Δt ,

$$y_i = \sum_{n=1}^M E_n \Phi_n(t_i - T_n) + \varepsilon_i, \quad 0 \leq i \leq N-1, \quad (1)$$

where $\{T_n, n \geq 1\}$ denotes the times of interaction between incoming photons and the detectors, $\{E_n, 1 \leq n \leq M\}$ is a sequence of independent and identically distributed (iid) random variables representing the photonic energies with unknown probability density function, $\{\Phi_n, 1 \leq n \leq M\}$ is a sequence of functions to be defined later which characterizes the electric pulse shapes generated by the photons, and $\{\varepsilon_i, 0 \leq i \leq N-1\}$ is a sequence of iid Gaussian random variables with zero mean and variance σ^2 representing the additional noise of the input signal. We assume that $\{T_n, n \geq 1\}$ is a sample path of a homogeneous Poisson process (HPP) with

unknown activity λ , denoted in the following sections by N . More specifically, given the interval A , $N(A)$ represents the number of electrical pulses whose starting times lie in A .

B. Debiasing for Sampled Signals Based on Poisson Process

Should the photon arrival times T_n be perfectly known, a known estimate of λ would be

$$\hat{\lambda}_{ideal} = \frac{M}{T_M}, \quad (2)$$

where M denotes the number of electrical pulses and T_M the arrival time of the last photon in the recorded signal. However, this estimate does not take into account the fact that the signal is sampled. This yields two different type of errors: an obvious rounding error, since the value of T_M in the previous estimate is rounded. But more importantly, we cannot derive from the sampled signal the exact number of impinging photons between two consecutive sampling times. This phenomenon introduces a bias in the aforementioned estimate which can lead to an underestimation of the activity, particularly if it is high.

In this paper, we suggest to rely on the knowledge of active bins, that is that we can retrieve efficiently the intervals between two sampling times (bins) when one or several photons interacted with the detector [11]. Denote by $\{B_n, 0 \leq n \leq N-1\}$ the sequence of random variables defined as follows:

$$B_n = \begin{cases} 1 & \text{if } N([t_n; t_{n+1})) > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Roughly speaking, B_n illustrates that between two sampling points we can infer whether an electrical activity started, but are unable to derive exactly how many photons caused it. It is well known (see e.g. [12]) that the random variables B_n are independent and identically distributed, with a common Bernoulli distribution with parameter $1 - \exp(-\lambda\Delta t)$. Consequently, a less biased estimate can be derived from the law of large numbers applied to $\{B_n, 1 \leq n \leq N\}$. More precisely, the estimate

$$\hat{\lambda} = -\frac{1}{\Delta t} \ln\left(1 - \frac{1}{N} \sum_{n=0}^{N-1} B_n\right) \quad (3)$$

takes into account the discretization of the signal and provides an unbiased estimate of the activity λ .

C. Activity Estimation Procedure

With these considerations in mind, it is possible to improve the method developed in [11] in order to make it more suitable for high levels of activity. As noted in [11], neither the energies nor the shapes are known. However, it is known that most detectors with excellent Signal to Noise Ratio (SNR), such as Germanium detectors, provide pulses characterized by a rapid growth created by a charge collection followed by an exponential decay as the charges migrate to the detector electrodes. Thus, a logical idea to represent the signal defined in (1) is to rely on some user predefined dictionary of truncated gamma shapes in order to obtain a reliable model. Since a

gamma shape is parametrized by two parameters — its shape and scale —, we define a set of p pairs of such parameters by $\theta = \{(\theta_1^{(s)}, \theta_2^{(s)}); s = 1, 2, \dots, p\}$. For all $s = 1, \dots, p$, we define the following pulse shapes:

$$\Gamma_s(t) = c_s t^{\theta_1^{(s)}} \exp(-\theta_2^{(s)} t) \mathbf{1}(0 < t \leq \tau\Delta t), \quad (4)$$

where τ is a positive constant integer defining the common support of the pulse shapes, $\mathbf{1}(0 < t \leq \tau\Delta t)$ is the indicator function, being equal to 1 when $0 < t \leq \tau\Delta t$ and to 0 otherwise, and c_s is a normalizing constant chosen so that $\frac{1}{N} \sum_{i=0}^{N-1} \Gamma_s(t_i)^2 = 1$. Accordingly, we define the following $N \times p$ matrix $\mathbf{A}_k$ whose columns are sampled versions of the previously defined pulse shapes, translated by t_k :

$$\mathbf{A}_k = [\Gamma_s(t_i - t_k)]_{0 \leq i \leq N-1, 1 \leq s \leq p}.$$

Intuitively, $\mathbf{A}_k$ can be understood as a subdictionary representing the electrical pulses starting from the same sampling time t_k . A global dictionary $\mathbf{A}$ of possible pulses occurring at different times is then defined by regrouping the $\mathbf{A}_k$'s altogether:

$$\mathbf{A} = [\mathbf{A}_0 \ \mathbf{A}_1 \ \dots \ \mathbf{A}_{N-1}].$$

Given any vector β of size pN , it is thus naturally decomposed by blocks of size p : $\beta = [\beta_0^T; \dots; \beta_{N-1}^T]^T$, where for all i , $\beta_i \in \mathbb{R}^p$. Thus, the model investigated throughout the rest of the paper is

$$\mathbf{y} = \mathbf{A}\beta + \varepsilon + \delta = \sum_{k=0}^{N-1} \mathbf{A}_k \beta_k + \varepsilon + \delta, \quad (5)$$

where $\mathbf{y} = [y_0, y_1, \dots, y_{N-1}]^T$ and $\varepsilon = [\varepsilon_0, \varepsilon_1, \dots, \varepsilon_{N-1}]^T$ represent the recorded signal and the additive noise respectively, β is a sparse non-negative vector whose non-null entries are related to photon arrival times, and $\delta = [\delta_0, \delta_1, \dots, \delta_{N-1}]^T$ is a vector quantifying the discretization error made between (1) and (5). Our objective is therefore to estimate λ from (5), and at first to find a sparse estimate of β , so that its non-zero entries provide relevant information on the active bins present in the signal. This sparse estimate is obtained by solving a non-negative version of the LASSO, that is:

$$\hat{\beta}(r) = \arg \min_{\beta \geq 0} \left\{ \|\mathbf{y} - \mathbf{A}\beta\|_2^2 + r \|\beta\|_1 \right\}. \quad (6)$$

Introduced in [13], LASSO provides a regressor $\hat{\beta}(r)$ whose sparsity is controlled by the parameter r (the larger, the sparser). In [11], theoretical results are presented which guarantee that the active bins can be related to the positive entries of (6), provided a threshold is applied on the regressor $\hat{\beta}(r)$. However, in [11] the number of active bins found were substituted directly in (2), yielding a bias in the activity estimation. In contrast, we suggest correcting this bias by using (3), as now described.

Given a sparsity parameter r , and assuming that a NN-LASSO regressor $\hat{\beta}(r) = [\hat{\beta}_0^T(r), \hat{\beta}_1^T(r), \dots, \hat{\beta}_{N-1}^T(r)]^T$ has been computed (we refer to [14] for a fast numerical method), we first estimate the active bins by a thresholding

of the blocks $\hat{\beta}_n(r)$, $n = 0 \dots N-1$. More specifically, we estimate the activity bins values as:

$$\hat{B}_n(r) = \mathbb{1}(\|\hat{\beta}_n(r)\|_1 \geq \eta), \quad (7)$$

where $\eta > 0$ is a user-specified threshold. The necessity for a threshold η as described in (7) can be understood from the properties of LASSO. It is well known (see e.g. [15] and references therein) that consistency in selection can be obtained only if some irrepresentable condition [16], involving well decorrelated columns in $\mathbf{A}$, holds. This is not the case here, since (1) is based on a dictionary with highly correlated functions. Therefore, it is most likely that (6) will select in $\hat{\beta}(r)$ sets of close coefficients to characterize one single pulse, even if the overall solution remains sparse. Furthermore, the sparsity parameter r is unknown, and a too low value of r could yield small coefficients in $\hat{\beta}(r)$, related to overfitting the noise. Therefore, only the most representative coefficients of $\hat{\beta}(r)$ must be selected, this being done by adding an additional thresholding step.

Eventually, the proposed estimator, that we shall call Unbiased LASSO Estimate, and shall denote by $\hat{\lambda}_{ULE}(r)$ in the following sections, is derived by substituting (7) in (3), namely

$$\hat{\lambda}_{ULE}(r) = -\frac{1}{\Delta t} \ln\left(1 - \frac{1}{N} \sum_{n=0}^{N-1} \hat{B}_n(r)\right). \quad (8)$$

As an additional remark, we can notice that when the activity is small (that is, when $\lambda \Delta t \ll 1$), the estimator (8) can be approximated by its first-order Taylor expansion $\hat{\lambda}_{ULE}(r) \approx \frac{1}{N \Delta t} \sum_{n=0}^{N-1} \hat{B}_n(r)$, which is equivalent to the estimate obtained in [11]. This illustrates the fact that for reasonable activities, $\hat{\lambda}_{ULE}$ behaves similarly to the estimator described in [11], and that we should expect similar performances, at least in this case.

III. RESULTS

We present in this section results on simulations and on a real dataset provided by CEA Saclay, from the PiDie experiment. LASSO regression was done using the SparseLab Matlab library.

A. Simulations Results

The experimental settings was chosen accordingly to [9]. We briefly recall them for convenience. The performances of the proposed approach are investigated for simulated signals stemming from a homogeneous Poisson process whose true intensity λ varies from 0.05 to 0.4. Those numbers are related to high or very high radioactive activities, as mentioned in [1]. The energies E_n are drawn according to a truncated Gaussian density, with mean of 100 and variance of 5. The noise variance is chosen equal to $\sigma = 1$, in order to reflect a good SNR which can be encountered in practice.

In order to check the robustness of the approach and its practical implementation for real-time instrumentation, the signals are simulated in two different settings:

- (I) for each point of the Poisson process, an electrical pulse shape is taken randomly from $\mathbf{A}$;

- (II) for each point of the Poisson process, a Gamma shape similar to (4) is created with parameters θ_1, θ_2 randomly chosen accordingly to a uniform distribution on $[0.3, 1.3]$.

20 different shapes (and their shifted versions) were used in the dictionary $\mathbf{A}$, with parameters $\theta_1 = \theta_2$ ranging from 0.4 to 1.2. This choice illustrates the fact that $\mathbf{A}$ must be carefully chosen, though some discrepancy between the shapes appearing in $\mathbf{y}$ and $\mathbf{A}$ may be allowed. It is obvious that the standard framework for regression is (I). However, in order to test the proposed approach on real world experiments, it is important to check how the algorithm behaves when the dictionary is not rich enough to cover all the possible shapes. Therefore, (II) should be investigated if we aim to propose fast algorithms for real-time implementations. For one given activity, we compare the performances of the proposed estimate $\hat{\lambda}_{ULE}(r)$ defined in (8) with the ones of three other estimates. The ideal estimate, $\hat{\lambda}_{ideal}$, is defined in (2), an estimate obtained by means of a peak detector is defined as

$$\hat{\lambda}_{peak} = \frac{\text{Number of detected peaks}}{\text{Location of the last peak detected}},$$

and the thresholded LASSO estimate described in [11], which is defined using the previous notations as

$$\hat{\lambda}_{LASSO}(r) = \frac{\sum_{n=0}^{N-1} B_n(r)}{\Delta t \max\{n; B_n(r) = 1\}}.$$

The activity estimator is computed 1000 times on signals $\mathbf{y}$ of size $N = 200$, both in cases (I) and (II).

The choice of the sparsity parameter r and thresholding parameter η is very important to obtain good results. As can be seen from the graph of the mean squared error (MSE) between the true activity value and its estimate in Figure 2, numerous (r, η) may provide satisfactory results. We found out experimentally that r should be set so that $\|\mathbf{y} - \mathbf{A}\beta(r)\|_2 \leq 3\sigma\sqrt{N}$ (the use of the LARS algorithm [14] makes this possible, thus circumventing the actual search of r), while setting η between 30 and 50 would provide satisfactory results. Finding a fully, systematic, data-driven choice of (r, η) remains as for now a challenging issue, and therefore is not in the scope of the present paper. The results obtained for $\eta = 50$ are displayed in Figure 3a and Figure 3b.

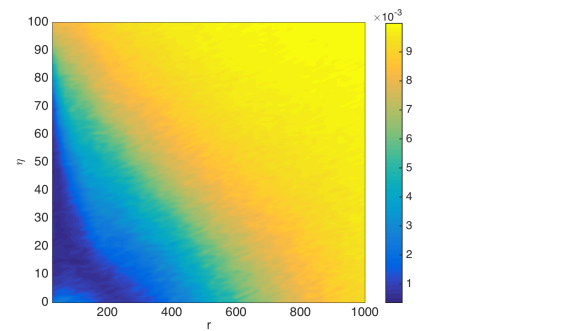
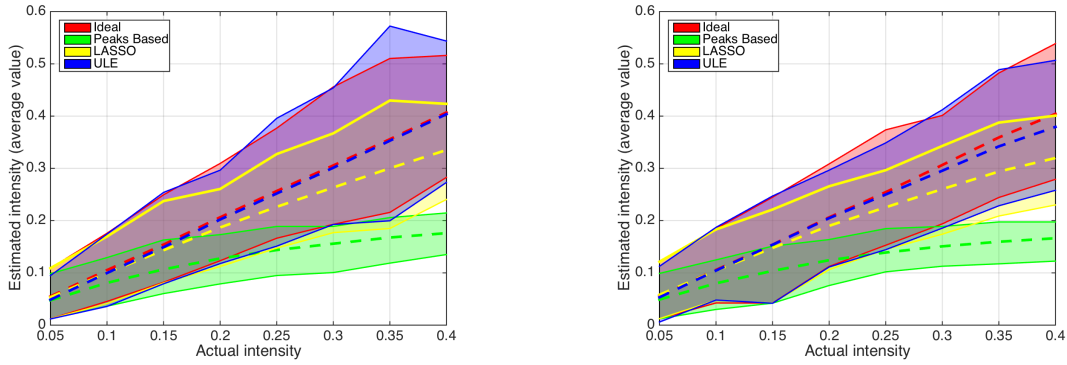


Fig. 2: Approximated MSE for $\lambda = 0.1$, case (I). The minimum is attained for $r = 50$ and $\eta = 28$.



(a) Mean values of the considered activity estimates, with 90% confidence intervals - case (I) (b) Mean values of the considered activity estimates, with 90% confidence intervals - case (II)

Fig. 3: Average of the estimated activities in the simulations, with confidence intervals. Red: ideal estimate. Green: peak-based estimate. Cyan: LASSO estimate. Blue: ULE estimate.

B. Results on Real Data

The proposed algorithm was tested on $N = 10^5$ points of a real dataset stemming from the PiDie experiment. In this framework, the real activity is 10^6 photons per second, which when reported to the sampling frequency of 10 MHz gives a normalized activity of 0.1. The signal investigated was split into sub-signals of size 100, each one providing a different estimate of the activity. r was set using the same rule as in the simulations, while we chose $\eta = 30$. It can be seen from Figure 4 that the arrival times can be well located. However, the average of the obtained activity estimates is 0.11, with a standard deviation of 0.01. This bias is related to Figure 3b, and illustrates that good fit between the recorded signal and $\mathbf{A}$ is required in order to obtain satisfactory results.

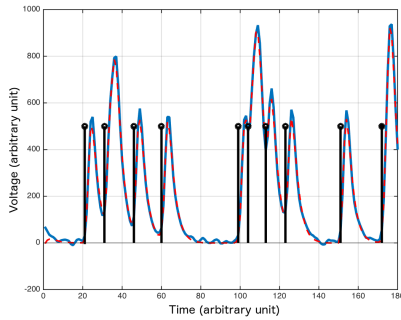


Fig. 4: Real signal (blue), estimated signal (dotted red) and arrival times (black)

C. Discussion

From the results obtained in the simulations, it can be seen that the proposed activity estimate performs better $\hat{\lambda}_{peak}$ and $\hat{\lambda}_{LASSO}$, particularly in cases of high levels of activity. More specifically, in the case of signals generated from $\mathbf{A}$, we can observe that the obtained values are close to the actual intensities, and that $\hat{\lambda}_{ULE}(r)$ performs similarly to the ideal estimate $\hat{\lambda}_{ideal}$ defined in (2). However, the case (II) is more

problematic, and the gain obtained by using $\hat{\lambda}_{ULE}(r)$ is less clear. This degradation of the performances illustrates the fact that a mismatch between the dictionary and the signal at hand can severely cripple the activity estimation. This point can be partially solved, either by calibrating the dictionary using a training set of pulse shapes, or by carefully setting the threshold parameter η . We can also remark that the sparsity parameter r and the thresholding parameter η depend on each other. However, there is also a less obvious dependency between these parameters and the photons energies E_n : should they be too low in average, r should be decreased and η increased to provide satisfactory results. A full discussion on the relation between η, r, E_n and recommendations on the choice of η and r can be found in [17, Propositions 2 and 3]. We may also notice that one of the inherent problems of LASSO is its poor performance in variable selection when dictionary shapes are very correlated. Indeed, LASSO tends to select a combination of close shapes to characterize one event, and to attribute a high coefficient to one of them randomly. This explains why a thresholding operation is required, particularly in the random setting. However, we are not interested in precise reconstruction of the signal itself, but rather in the number of arrival times. Therefore, this random attribution is of mild consequence, provided the selected variable remains close to the actual arrival time.

IV. CONCLUSION

We have presented an improvement of an existing algorithm for the estimation of the activity of homogeneous Poisson processes, based on a recorded, digital signal. Results illustrate that though the debiasing formula does improve the estimate of the activity, it is necessary to guarantee that the dictionary used is well suited to the data needed to process. In case of partial fit, we can observe a degradation of the performances, though these performances remain better than the already existing method. Future contributions will provide theoretical results on the presented approach and further insights such as confidence bounds around selected coefficients.

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